

# MoSPI — AI-Based Project Monitoring Platform

Full English Version of the Project Notes

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## Project Statement — SIH26103

MoSPI has a large amount of historical data related to major infrastructure projects, including project costs, expenditure, physical progress, milestones, timelines, delays and revisions.

Currently, this data is mainly used for monitoring, reporting and generating charts, but its historical patterns are not fully utilized to identify risks in ongoing projects.

Our proposed solution will use Machine Learning to learn patterns from completed and historical projects and apply those patterns to ongoing projects. The system will predict the probability of cost overruns and time delays, identify the major factors contributing to risk, assign an overall project-risk score, and generate early warnings when a project starts showing a pattern associated with future problems.

**The objective is to move MoSPI from reactive monitoring — “What has already happened?” — to proactive monitoring — “What is likely to happen next, and which project needs attention now?”**

## Main Problems Identified

- Project's actual progress should be known accurately and on time.
- Delays are often identified only after they have already happened.
- Cost Overrun.
- Time Overrun.
- Comparing different projects is difficult.
- The actual reason behind a problem is difficult to identify.
- The huge amount of historical data is not being fully utilized.
- Officers have to decide priority among thousands of projects.

# 1. Existing Situation / Problem Faced

Project or site information does not reach the concerned authorities daily in an exact and accurate manner. Information often reaches them late, and by the time it is provided through officers, the work may already have progressed or been completed.

## Problems

The notes identify a risk of contractors and officers manipulating reported quantities, management information and bills. If the reported amount or quantity is increased beyond actual work, the project cost can increase and the government can suffer financial loss.

## Persons / Roles Involved

| Role           | Main Responsibility                                                       |
|----------------|---------------------------------------------------------------------------|
| Contractor     | Executes the project or assigned work package.                            |
| Site Engineer  | Supervises site work and handles measurement/billing-related information. |
| AE / AM        | Assistant-level management role.                                          |
| GM             | General Manager.                                                          |
| PM             | Project Manager.                                                          |
| MCE            | Material Engineer.                                                        |
| Sub Engineer   | Visits the site for measurement and checking.                             |
| QA/QC Engineer | Quality Assurance / Quality Control and construction-quality checking.    |

## Construction Project Hierarchy — Top to Bottom

- Client / Owner → PMC → Project Director / Head → GM → PM → AM
- Construction Manager → Project Engineer → Site Engineer → Sub Engineer
- QA/QC Engineer → Planning Engineer → Billing Engineer → Safety Engineer / HSE
- Surveyor → Supervisor → Foreman → Skilled Workers → Labour / Helpers

The detailed hierarchy includes responsibilities such as budget and approvals at the client level, project management and execution at management levels, site supervision and measurements at engineering levels, and actual construction work at the worker/labour level.

## 2. Contractor & Petty Contractor Workflow

Planning Engineer — prepares the project schedule.

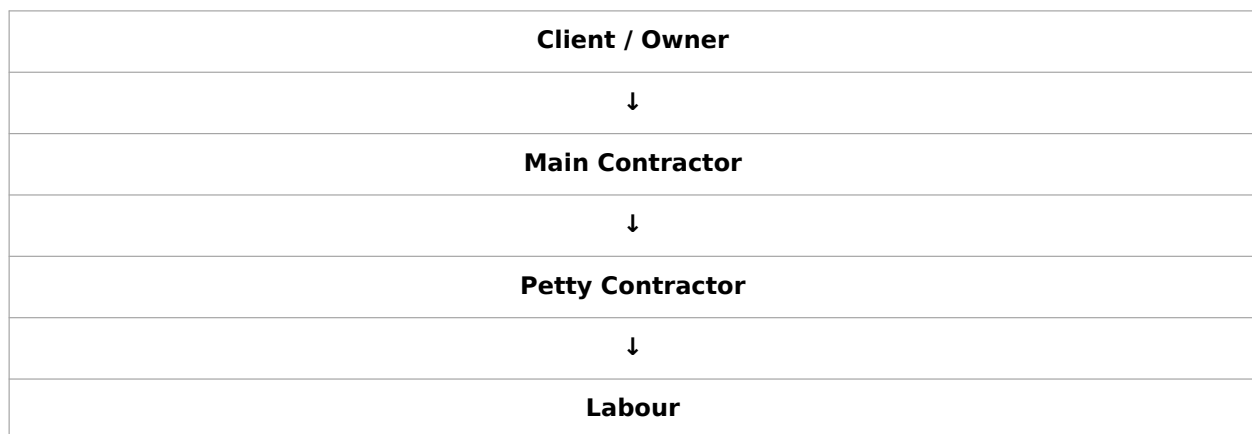
Billing Engineer — prepares and processes bills.

When work starts at a site, the contractor who has received the government tender first checks the resources required to complete the project.

If all resources are available, the contractor can complete the project directly. The second option is to assign a specific work package to a petty contractor.

### Petty Contractor

**A petty contractor is a contractor working under the main contractor who executes a small or specific construction work package.**



Example: A main contractor receives an entire building project and gives the brickwork package to a smaller contractor.

### A petty contractor may:

- Arrange masons and labour.
- Carry out the assigned brickwork or other package.
- Supervise daily work.
- Receive payment according to the assigned quantity/work.

### Important Difference

Labour performs physical work only. A petty contractor arranges/manages labour and takes responsibility for completing a particular work or package.

### The contractor checks resources and availability and decides the type of contract.

- Material + Tool + Worker + Maintenance.
- Material + Worker + Tool.
- Material + Tool.
- Tool + Worker.

As per the work requirement, the contractor decides how many workers, how much material and how many tools will be used, along with the required quantity and quality.

A petty contractor normally receives payment according to the work performed. However, where the petty contractor is responsible for material, tools and workers, the notes identify a possible risk that material may be diverted/sold, additional money may be demanded, or the full quantity of material may

not be used at the site.

## Main Contractor Risk

The notes identify a situation where, for higher profit, the main contractor may influence site engineers, material engineers, billing engineers, sub engineers and QA/QC personnel.

If quality checks are incorrectly performed or quantities are incorrectly reported, the bill may be increased beyond the actual work. Such records may then be submitted to the government, creating a financial loss and increasing project cost.

## Proposed Solution

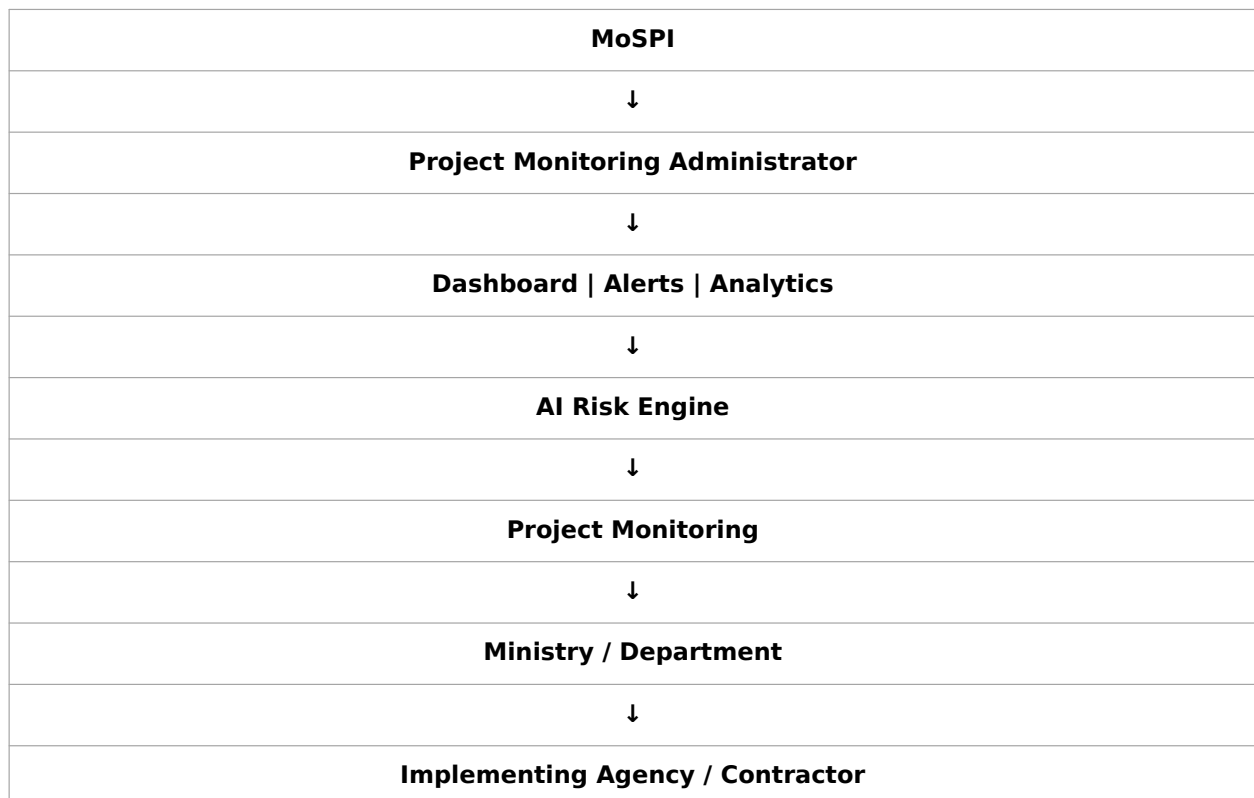
The proposed solution is an AI automation application that maintains a daily record from project start to project completion. The objective is to provide continuous, evidence-based visibility into what is happening at the site.

A dedicated Daily Record page will contain day-wise information, with a filter option so the monitoring team can select a date and see what happened on that day.

### Project details entered at the beginning may include:

- Project/work type
- Size
- Location
- Cost
- Duration
- Contractor
- Other required project information

## High-Level System Flow



### **3. Daily Site Monitoring & Evidence Collection**

All relevant officers can be connected to the project, while administrative control remains with the authorized MoSPI team.

#### **Day 1 — First Site Photo**

The supervisor uploads a site-location photo. The AI analyzes it and identifies the starting condition. It can ask recommended questions such as what happened on Day 1 and how much time it took. These questions can be optional. The pre-construction condition is added to the Daily Record.

#### **Day 2 — Material Unloading**

The uploaded photo, material quantity and work status are recorded. The AI can ask whether the expected work for that day was completed because it knows the planned workflow. Every material bill can be uploaded and later verified by the inspection team during a site visit.

The AI can ask how much material was received and request the corresponding bill/receipt. This information becomes part of the project record.

#### **Day 3 — Site Work Starts**

The status can be shown as Running. The supervisor/site team uploads photos from multiple angles, reports how many metres of work were completed, records the material used, and uploads supporting material photographs.

For example, cement bags or other consumed material can be recorded so that quantity can be cross-checked. Labour information can also be captured. Machinery fuel and rent expenses may be optional.

The Billing Engineer verifies bill-related information, while the system maintains the complete digital trail.

## **4. Continuous Monitoring of Officers**

The same process continues until the final day of the project.

The system connects the relevant officers and sends reminders according to their assigned responsibilities. For example, Material Engineer and Quality Engineer tasks can generate reminders when their involvement is required.

The Site Engineer visits the site and uploads a geo-tagged photo along with work photographs in the Daily Record. This provides evidence that the site visit took place.

Whenever the Sub Engineer has an assigned role, the officer visits the site, performs the inspection and uploads a geo-tagged photo/location.

The same mechanism applies to other relevant officers. The MoSPI team can continuously check site visits. If an officer has not completed a required visit, the system generates a reminder or signal to the monitoring team and the concerned officer.

## **5. Documentation Page**

A separate Documentation page can be created where officers submit project-related documents. The MoSPI team reviews the submitted documents, and access to this section remains under the authorized monitoring team.

Each officer can submit the required documents and maintain a complete digital record. This creates a traceable history of who visited the site, when they visited, and what evidence or documents they submitted.

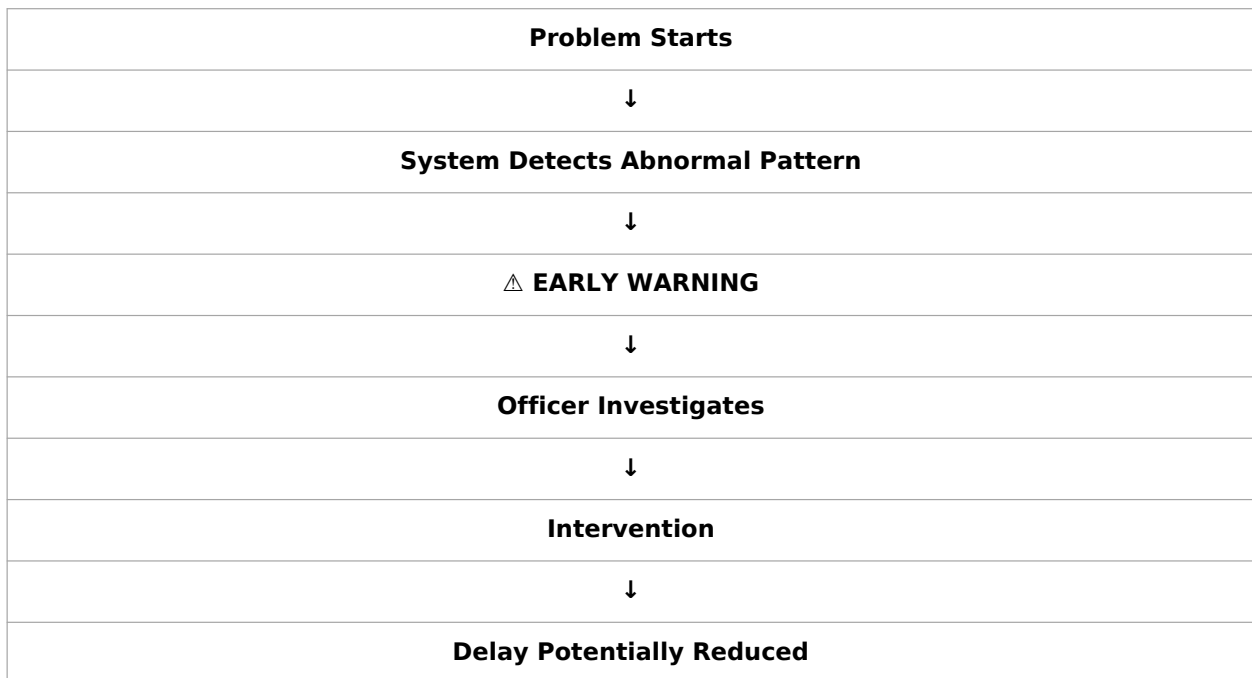
## 6. Delay Detection — Problem 2

The most important objective is to identify a developing delay before it becomes a major officially reported delay.

### Traditional Monitoring



### Proactive Monitoring with AI



**The objective is to convert after-the-fact reporting into an early-warning system.**

### Common Reasons for Delay

- Workers are not available.
- Workers do not report to the site.
- Required tools are unavailable.
- Materials/resources are unavailable.
- Land or legal issues affect the site.
- A local land-related issue or stay order stops the work.

- Machinery is not available on time.

The Daily Record will contain the reasons for delay, allowing the AI to generate reports and signals for officers.

The system can ask the site team for a daily update: if the site is ON, upload photos and status; if the site is OFF, provide the reason. The complete project status can then be viewed by the MoSPI team and shown in project statistics.

## 7. Cost Overrun — Problem 3

Infrastructure projects can have a difference between the original sanctioned cost and the final/revised cost.

|                                               |
|-----------------------------------------------|
| <b>Original Cost — ₹500 Cr</b>                |
| ↓                                             |
| <b>Scope / Material / Land / Other Issues</b> |
| ↓                                             |
| <b>Revised Cost — ₹650 Cr</b>                 |

**The important question is not only: “Cost has become ₹650 Cr.” The better question is: “When did the risk of increasing from ₹500 Cr to ₹650 Cr become visible?”**

Another important question is: “Which factors were early indicators of cost escalation?”

### Cost-Monitoring Solution

Material data, bills, labour information, machinery information and other daily records will be stored in the system. The AI can generate a cost report and flag unusually high billing.

If a Billing Engineer enters a higher amount, the system can ask for the reason. The authorized MoSPI team can then review and approve the submission.

If a Billing Engineer submits a bill representing 20% progress but the quantity appears incorrect, the AI can compare the bill with the Daily Record and request a reason before approval.

The AI can use construction-related knowledge to calculate an expected quantity/material requirement and compare it with the submitted bill. A significant mismatch can generate an alert.

All such information can be cross-checked by the monitoring team. Documentation submissions can also be checked against the Daily Record.

Expected outcome: the opportunity for corruption or incorrect billing is reduced through continuous records, cross-checking and controlled approval.

## 8. Time Overrun — Problem 4

The same principle applies to project completion dates.

|                                         |
|-----------------------------------------|
| <b>Original Completion — March 2025</b> |
| ↓                                       |
| <b>Extension — September 2025</b>       |
| ↓                                       |
| <b>Further Extension — June 2026</b>    |

**Ideally, the government should receive an earlier signal such as: “Based on the current trajectory, completion by March 2025 is unrealistic.”**

The automation continuously analyzes progress against the deadline. When a new project is created and progress changes are recorded, the system estimates whether the current trajectory is consistent with the planned completion date.

If work is not progressing according to the deadline, a warning/reminder is sent to the relevant officers.

A key feature is that the manager can see the reason behind the delay on a daily basis.

## 9. Different Projects — Comparison Problem

Suppose there are 50 highway projects, 20 railway projects, 15 bridge projects and 30 power projects.

Every project may have a different size, location, cost, duration, contractor, implementing agency and level of complexity.

Therefore, simply saying “Project A is 70% complete” is not always meaningful.

**A better comparison is: “Project A is 70% complete, but comparable projects at this stage average 82%.” This is where benchmarking becomes useful.**

## 10. Multi-Site Project Comparison

The application will provide multi-site management so a manager can view different projects from one place. Projects with slower progress can be highlighted by the AI.

A Comparison page can show deadline, progress, cost, contractor and other relevant details. Statistics can identify which projects are slower and which are progressing better.

The comparison information will use the Daily Record and documentation/payment information. The view is intended for monitoring/statistics and should not provide an editing option for project records.

### **The project statistics should show information such as:**

- Progress
- Deadline
- Cost
- Contractor
- Amount paid
- Project status
- Relative performance compared with similar projects

## 11. Identifying the Actual Reason Behind a Problem — Problem 6

**A dashboard can easily show “Project Delayed.” However, the officer needs to know: Why?**

### **Possible reasons include:**

- Land acquisition
- Forest/environmental clearance
- Utility shifting
- Contractor performance
- Tender issues
- Funding/payment issues
- Design changes
- Scope changes
- Material availability
- Geological conditions
- Weather
- Law and order
- Local issues
- Inter-department coordination
- Litigation

## 12. Root-Cause Identification & Responsibility

Reason identification should be an important part of the solution.

If a problem matches the daily monitoring data and recorded reasons, it can move into the Reason section. This section should explain what happened, why it happened and how it can be addressed.

For example, if material is unavailable, the Material Engineer can receive a signal to arrange/order the material. The reason and action can then be reviewed by the upper officer.

If funding is not received on time, the system can generate a notice and assign the relevant responsibility, including to the Billing Engineer where appropriate.

The MoSPI team can monitor these actions. The Site Manager can also check daily officer assignments and working status.

This reduces the need for repeated calls and helps the team identify the actual reason for a problem directly on the platform.

If a reason is assigned but the responsible officer does not respond, the system generates another signal and keeps a record of the time, date and purpose of the notification.

## 13. Historical Data — Problems 7 & 8

Problem 7: What is happening to the historical data now?

Problem 8: How can the same data be used better?

### What is Historical Data?

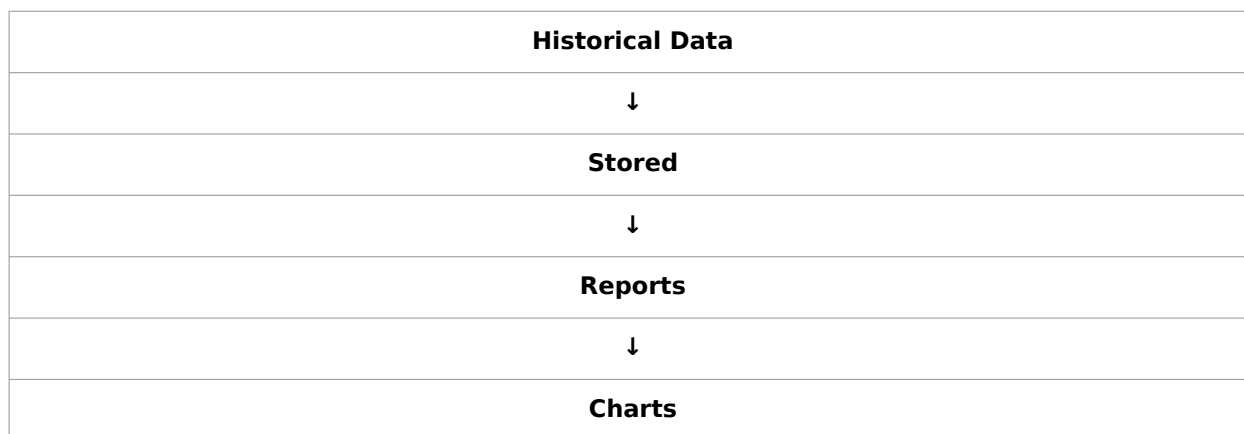
MoSPI has data from old infrastructure projects. For example, a road project may contain a five-year record of planned progress, actual progress and expenditure.

## 14. Example of Historical Project Data

| Year | Planned Progress | Actual Progress | Expenditure |
|------|------------------|-----------------|-------------|
| 2022 | 20%              | 15%             | ₹100 Cr     |
| 2023 | 40%              | 31%             | ₹250 Cr     |
| 2024 | 65%              | 48%             | ₹430 Cr     |
| 2025 | 85%              | 62%             | ₹620 Cr     |
| 2026 | 100%             | 78%             | ₹850 Cr     |

This five-year record is historical data. Additional information can include original project cost, revised cost, start date, scheduled completion, actual completion, physical progress, financial progress, milestones, delays, extensions, project status, ministry and sector.

### Current Basic Use of Historical Data



Simple meaning: data is collected → stored in the database → used to create reports → used to create charts.

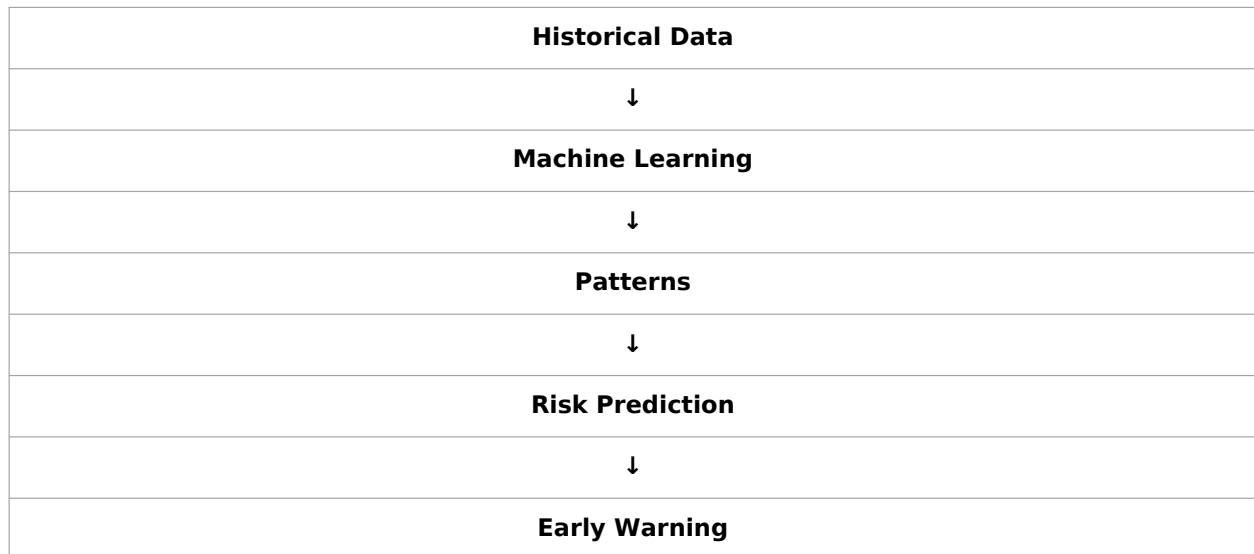
For example, an officer dashboard may show: Project A, Physical Progress = 78%, Expenditure = ₹850 Cr, Status = Ongoing.

A report may show 2022 → 15%, 2023 → 31%, 2024 → 48%, 2025 → 62%, 2026 → 78%, followed by a graph.

**The problem is not that the data is bad. The problem is that the system primarily answers: “WHAT HAS HAPPENED?” The administrator also needs: “WHAT IS LIKELY TO HAPPEN NEXT?”**

## 15. Using Historical Data for Machine Learning

The same historical data can be used as training material for Machine Learning.



Assume there are 10,000 completed/past projects. ML examines these projects and learns patterns.

For example, suppose the model identifies that projects where expenditure increased very quickly, physical progress remained continuously below expectations, and milestones were repeatedly delayed had a higher historical probability of future cost/time overrun.

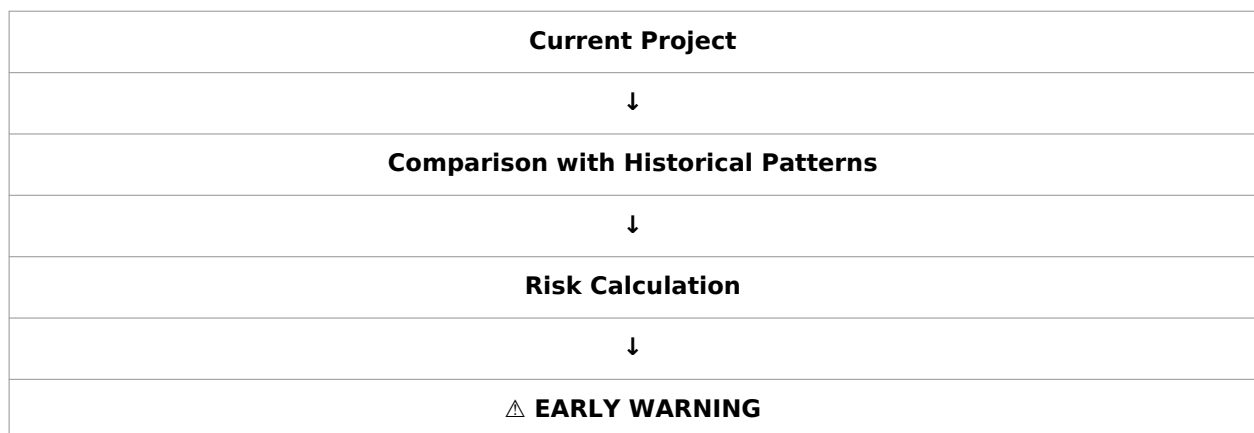
**That combination becomes a learned pattern.**

### Applying the Model to a Current Project

Suppose the current Project X has:

- Expected Progress = 70%
- Actual Progress = 52%
- Expenditure = 68%
- Milestones = 3 delayed

A human officer may simply say: “The project appears to be slow.” ML can compare the current progress-expenditure pattern with historical projects and state that projects with a similar pattern historically had a high probability of schedule overrun.



## 16. Meaning of Risk Prediction

Risk prediction does not mean predicting the future with 100% certainty.

The system should not say: “The project will definitely be 8 months late.” Instead, it can say: “Based on current data, the estimated probability of the project being delayed by 8+ months is 82%.”

### Example Risk Output

| Project X         | Risk                |
|-------------------|---------------------|
| Cost Overrun Risk | 74%                 |
| Time Overrun Risk | 86%                 |
| Overall Risk      | 89 / 100 — CRITICAL |

## 17. Early Warning — Main Benefit

Suppose a project is still officially marked as “Ongoing” and a delay has not yet been declared. The ML model may still identify an increasing risk trend.

| Month     | Risk Score |
|-----------|------------|
| April     | 38         |
| May       | 44         |
| June      | 51         |
| July      | 63         |
| August    | 78         |
| September | 87         |

**The system can alert the officer: “Early Warning: Project risk has increased significantly.”**

### Primary indicators may include:

- Physical progress is below the planned trajectory.
- Expenditure is progressing faster than physical work.
- Multiple milestones are delayed.

The officer can investigate the problem before it becomes officially severe.

## 18. Real-Life Highway Project Example

Assume a highway project with:

- Project Cost = ₹1,000 Cr
- Duration = 3 years

### First Year

- Expected physical progress = 30%
- Actual physical progress = 24%
- Expenditure = 32%

At this stage, there may not appear to be a massive problem.

### Second Year

- Expected progress = 65%
- Actual progress = 47%
- Expenditure = 68%

**Here there is an important signal: 68% of the money has been spent, while physical work is only 47% complete.**

Historical data may show the ML model that projects with such patterns have historically had a higher risk of future cost/time overrun.

| Current Data |
|--------------|
| ↓            |

|                                            |
|--------------------------------------------|
| <b>Comparison with Historical Projects</b> |
| ↓                                          |
| <b>Pattern Detected</b>                    |
| ↓                                          |
| <b>High Risk</b>                           |
| ↓                                          |
| <b>⚠️ EARLY WARNING</b>                    |

## 19. Important Point About Historical Data

The system should not rely only on historical data for prediction. Historical data provides the learning material.

### Think of it like this:

If a person has observed 1,000 road projects, that person may use experience to say: “This type of pattern generally creates problems.” ML performs a similar pattern-learning task, but it can analyze many variables simultaneously.



## 20. Overall Proposed Platform

The complete concept combines continuous daily site records, document management, officer/site-visit verification, cost and time monitoring, project comparison, root-cause identification and Machine Learning-based risk prediction.

**The intended outcome is a shift from delayed, report-based monitoring to continuous, evidence-based and proactive project monitoring, where the system can highlight emerging risks before they become major problems.**

### Final Concept in One Flow



|                                     |
|-------------------------------------|
| <b>Root Cause Identification</b>    |
| ↓                                   |
| <b>Risk Score + Early Warning</b>   |
| ↓                                   |
| <b>Officer / MoSPI Intervention</b> |

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End of Full English Version